Do You Know?

- Electricity travels at the speed of light - more than 186,000 miles per second!
- Electricity always tries to find the easiest and shortest path to the ground.
- Electricity can be made from wind, water, the sun and even animal poop.
- A 600-megawatt natural gas plant can power 220,000 homes.
- The first power plant - owned by Thomas Edison - opened in New York City in 1882.

5.5 THREE PHASE STAR/ DELTA CONNECTIONS

3 phase A.C generator is shown in Fig 5.13. Three identical windings A,B and C are placed 1200 electrical degree apart. It rotates in anti-clock wise direction. It generates 3 phase supply. Three phases are indicated in Red (R), Yellow (Y) and Blue (B) colors.

5.5.1 Poly Phase System

If the armature of an alternator generating AC apply is having only one winding, it generates single phase supply. Instead of one winding, if the alternator has two or three windings, then two or three phases are generated respectively. So a system produces more than one phase is called poly-phase system.

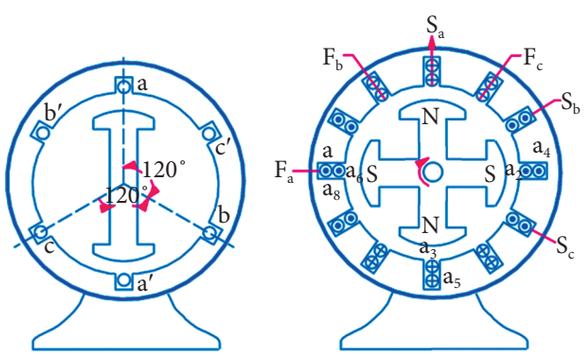


Fig 5.13 *Poly Phase System Generation*

5.5.2. Reasons for the Use of Three-Phase System

Three phase system is preferred over single phase system for the following reasons.

i. Three phase power has a constant magnitude whereas single phase power pulsates from zero to peak value at twice the supply frequency.

ii. A three phase system can set up a rotating magnetic field in stationary windings. This cannot be done with a single phase current.

iii. For the same rating, Three phase machines (generators, motors, transformers) are smaller, simpler in construction and have better operation than single phase machines.

iv. To transmit the same amount of power to a fixed distance at a given voltage. The three phase system requires only three-fourth weight of copper that is required by the single phase system.

v. The voltage regulation of a three phase transmission line is better than that of a single phase line.

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